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AVEVA WORLD

CONFERENCE

Accelerate Innovation: Transform Your Team's Future

Predictive Analytics Deep Dive

Session ID: AP-04

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2019-11-13

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Overview of Predictive Analytics

Predictive Analytics Process

Demo

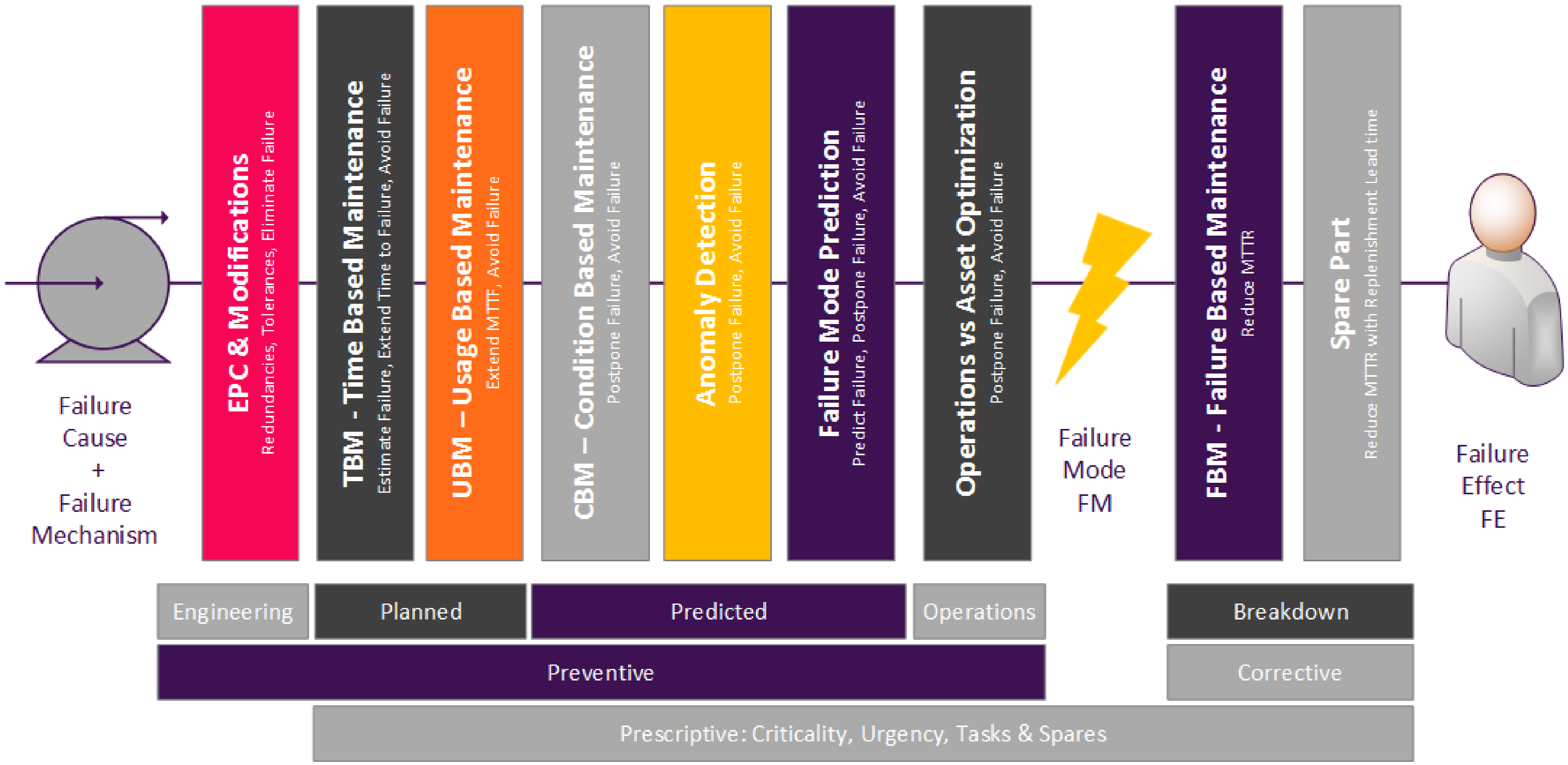
Architecture

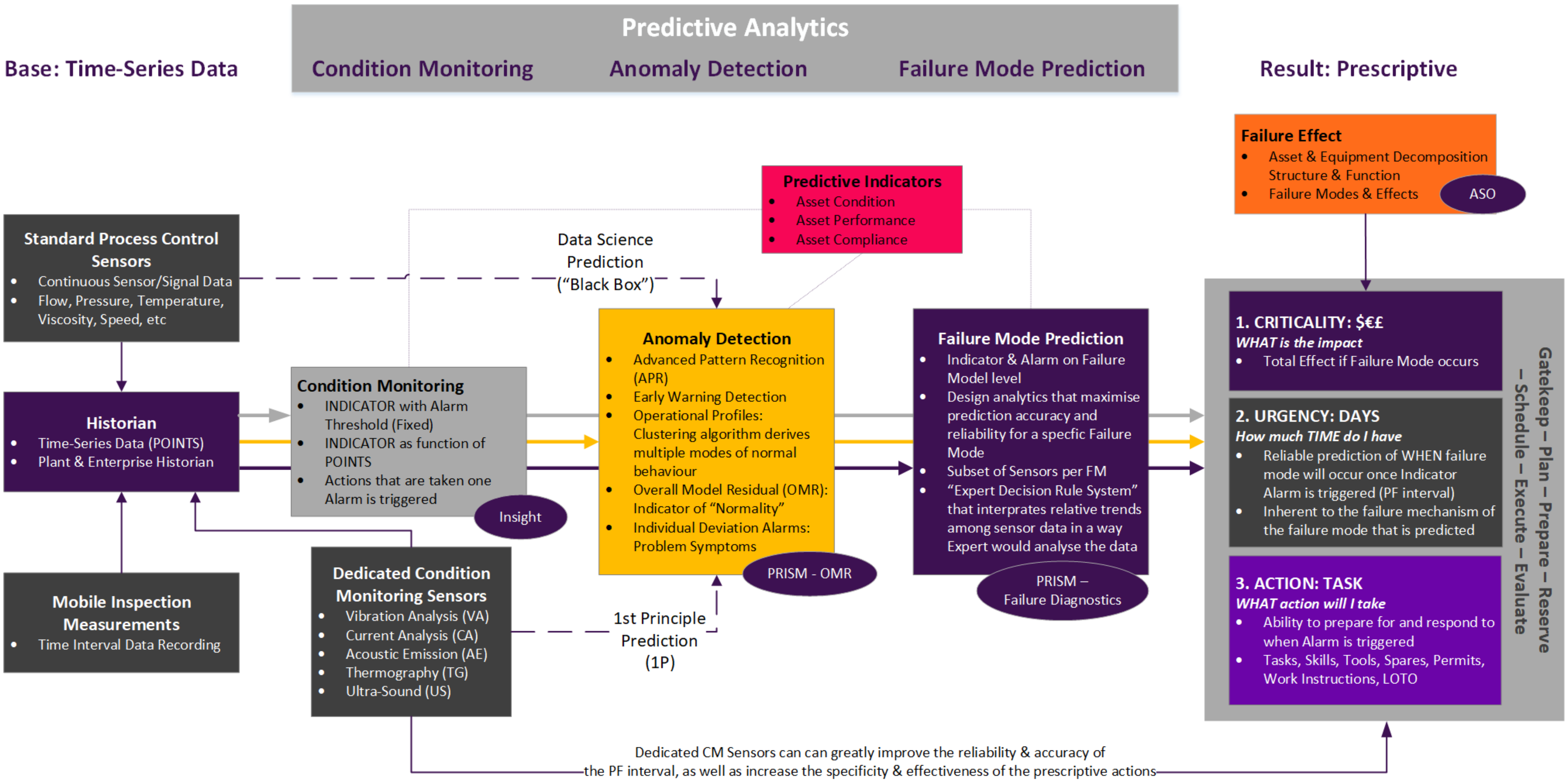
Roadmap

Predictive Analytics in the Cloud

Asset Strategy Optimization

Preventive vs Corrective Strategies that mitigate Failures to an acceptable level




Predictive Analytics
Base: Time-Series Data
Condition Monitoring
Anomaly Detection
Failure Mode Prediction
Result: Prescriptive
Failure Effect
Asset & Equipment Decomposition Structure & Function
Failure Modes & Effects
ASO
Predictive Indicators
Asset Condition
Asset Performance
Asset Compliance
Standard Process Control Sensors
Continuous Sensor/Signal Data
Flow, Pressure, Temperature, Viscosity, Speed, etc
Data Science Prediction (“Black Box”)
Historian
Time-Series Data (POINTS)
Plant & Enterprise Historian
Condition Monitoring
INDICATOR with Alarm Threshold (Fixed)
INDICATOR as function of POINTS
Actions that are taken one Alarm is triggered
Insight
Anomaly Detection
Advanced Pattern Recognition (APR)
Early Warning Detection
Operational Profiles: Clustering algorithm derives multiple modes of normal behaviour
Overall Model Residual (OMR): Indicator of “Normality”
Individual Deviation Alarms: Problem Symptoms
PRISM - OMR
Failure Mode Prediction
Indicator & Alarm on Failure Model level
Design analytics that maximise prediction accuracy and reliability for a specfic Failure Mode
Subset of Sensors per FM
“Expert Decision Rule System” that interpretes relative trends among sensor data in a way Expert would analyse the data
PRISM – Failure Diagnostics
1. CRITICALITY: $€£
WHAT is the impact
Total Effect if Failure Mode occurs
2. URGENCY: DAYS
How much TIME do I have
Reliable prediction of WHEN failure mode will occur once Indicator Alarm is triggered (PF interval)
Inherent to the failure mechanism of the failure mode that is predicted
3. ACTION: TASK
WHAT action will I take
Ability to prepare for and respond to when Alarm is triggered
Tasks, Skills, Tools, Spares, Permits, Work Instructions, LOTO
Gatekeep – Plan – Prepare – Reserve – Schedule – Execute – Evaluate
Mobile Inspection Measurements
Time Interval Data Recording
Dedicated Condition Monitoring Sensors
Vibration Analysis (VA)
Current Analysis (CA)
Acoustic Emission (AE)
Thermography (TG)
Ultra-Sound (US)
1st Principle Prediction (1P)
Dedicated CM Sensors can can greatly improve the reliability & accuracy of the PF interval, as well as increase the specificity & effectiveness of the prescriptive actions

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Predictive Analytics

What is it, and when to use it.

- Based on historical data and current data, make predictions about future events
- Compares current conditions to historical patterns to identify anomalies
- Can be complemented with Prescriptive, Prognostics, Remaining Useful Life
- Use predictive analytics when:
  - Asset is critical to operations, safety, has high revenue/cost impact
  - Have sufficient instrumentation (>= 3 sensors) and historical data
  - First principles, or Condition-based maintenance are not possible. Could be in combination.
  - There is sufficient lead time to take actions to minimize impact of failure

Predictive Analytics

Typical process

Discovery

- Identify critical assets and related data points
- Identify normal operating conditions in historical data
- Use or create templates for asset type and functional models

Model Building

- Cleanse data and set model running conditions
- Create and deploy Operational Profile
- Fine-tune Models

Monitoring

- Monitor anomaly alerts
- Diagnose alerts and identify potential failure mode. Document in Case Management
- Act: notify plant maintenance, adjust/retrain model

Discovery

- Identify critical assets and related data points
- Identify normal operating conditions in historical data
- Use or create templates for asset type and functional models

- Work with plant management, process/maintenance engineers, safety to identify critical assets. For example: No redundancy / single point of failure, No spares, Costly to repair, How often it runs
- Leverage P&IDs, historians to identify related points to asset. Sensors may be outside perimeter of asset, ambient conditions, production-related, etc. Points must be historized.
- Work with production personnel to identify periods of good operation
- Template Design Tool are Excel spreadsheets identifying model structure (group related tags), alerts limits, fault diagnostics, and prescriptive information. Add data point identifiers for each asset that will leverage the template.

Templates

Guidelines

- Complicated equipment may need to be split up into multiple models
  - Pump: Mechanical, Process
  - Gas Turbine: Compressor, Combustion, Turbine Cooling, Mechanical (Bearing Temperatures), Mechanical (Bearing Vibrations)
  - Centrifugal Compressor: Process (one model per stage), Seal System, Mechanical (Bearing Temperatures), Mechanical (Bearing Vibrations)
- Create separate models for the "driver" equipment as well
  - Ex: Motor → Gearbox → Compressor
- Rule of Thumb: Aim for ~10-25 tags per model
- Remember, not every tag has to be used in the Operational Profile
  - Use Actual Value alarms on tags that are important to monitor but don't "relate" well to anything else

Template Design Tool (TDT)

Asset Type

Template Description

Template description, constraints, etc.

Revision History

Revision	Date	Author	Description
0			First Issue

Contact Information

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For support information, contact eDNA Support

Model Building

- Cleanse data and set model running conditions
- Create and deploy Operational Profile
- Fine-tune Models

- Projects are collections of training data sets, operational profiles, filters, alerts definitions and notes. Projects are organized in an asset hierarchy.
- Import historical data
  - Rule of Thumb: 1 year at 1 hour intervals
- Data Cleansing:
  - Remove invalid data points (bad data, outliers, incomplete sets, etc.)
  - Remove periods of failure
  - Remove startup/shutdown periods
- Identify model running conditions:
  - Create filters identifying when asset is running
- Deploy Operational profile
  - Create Operational Profile (leave default advanced parameters)
  - Deploy Operational Profile (one profile per project)
- Fine-tune Models
  - Use Data Playback with previous failures and validate that model would have detected them
  - Typically need a month to fine-tune the model

Monitoring

- Monitor anomaly alerts
- Diagnose alerts and identify potential failure mode. Document in Case Management
- Act: notify plant maintenance, adjust/retrain model

- Runtime anomalies are organized in alerts in PRiSM Web.
- Prioritize alerts with User Lists, Grid sorting and filtering
- Diagnosis of an alert starts with trending of Overal Model Residual (OMR) and model signals.
- Sensor contribution identifies which signals contributes the most to OMR.
- Fault diagnostics identifies potential failure modes, along with prescriptive actions.
- Based on diagnosis of alert:
  - Notify plant maintenance
  - Retrain/adjust model
  - Wait and monitor

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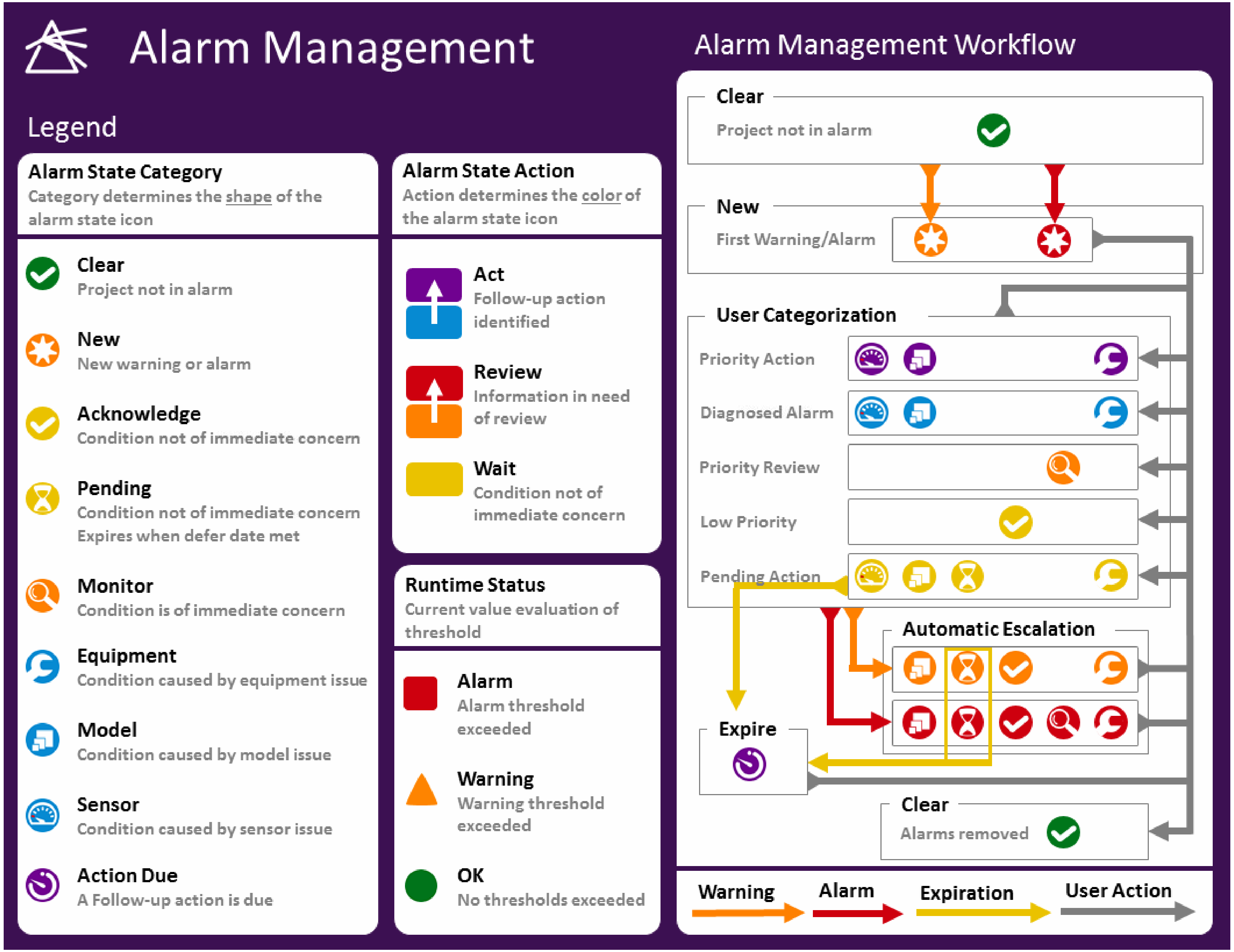
Alert Management

Workflow



Monitoring process

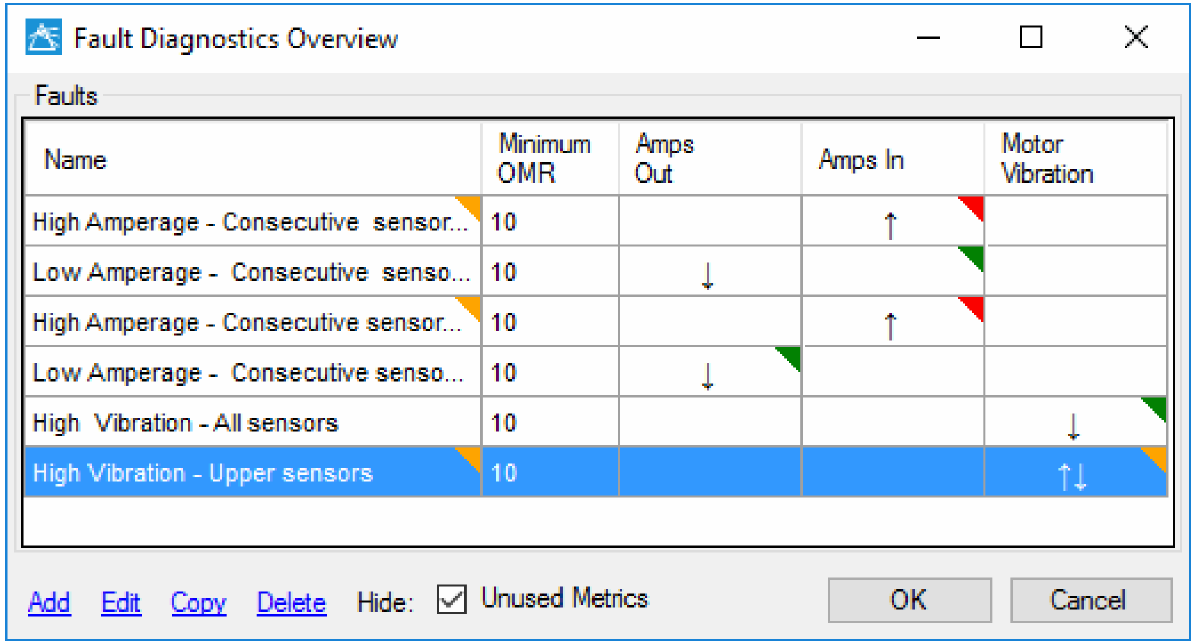
Developping a rythm

- A regular process helps set and maintain expectations between the monitoring center and the sites
- Goal is to provide early warning of days, weeks
- Not a replacement for operations

Fault Diagnostics

Design philosophy

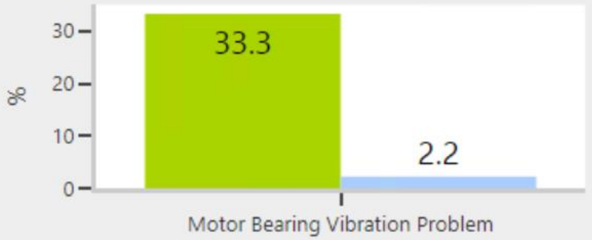
- Create fault diagnostics based on your equipment knowledge
  - Use them as a way to store expert knowledge in the software
- Think about what failures you've seen on this equipment in the past
- Fault diagnostics don't need to be added immediately
  - A well-tuned model with no fault diagnostics is better than a poorly-tuned with great fault diagnostics



Fault Diagnostics

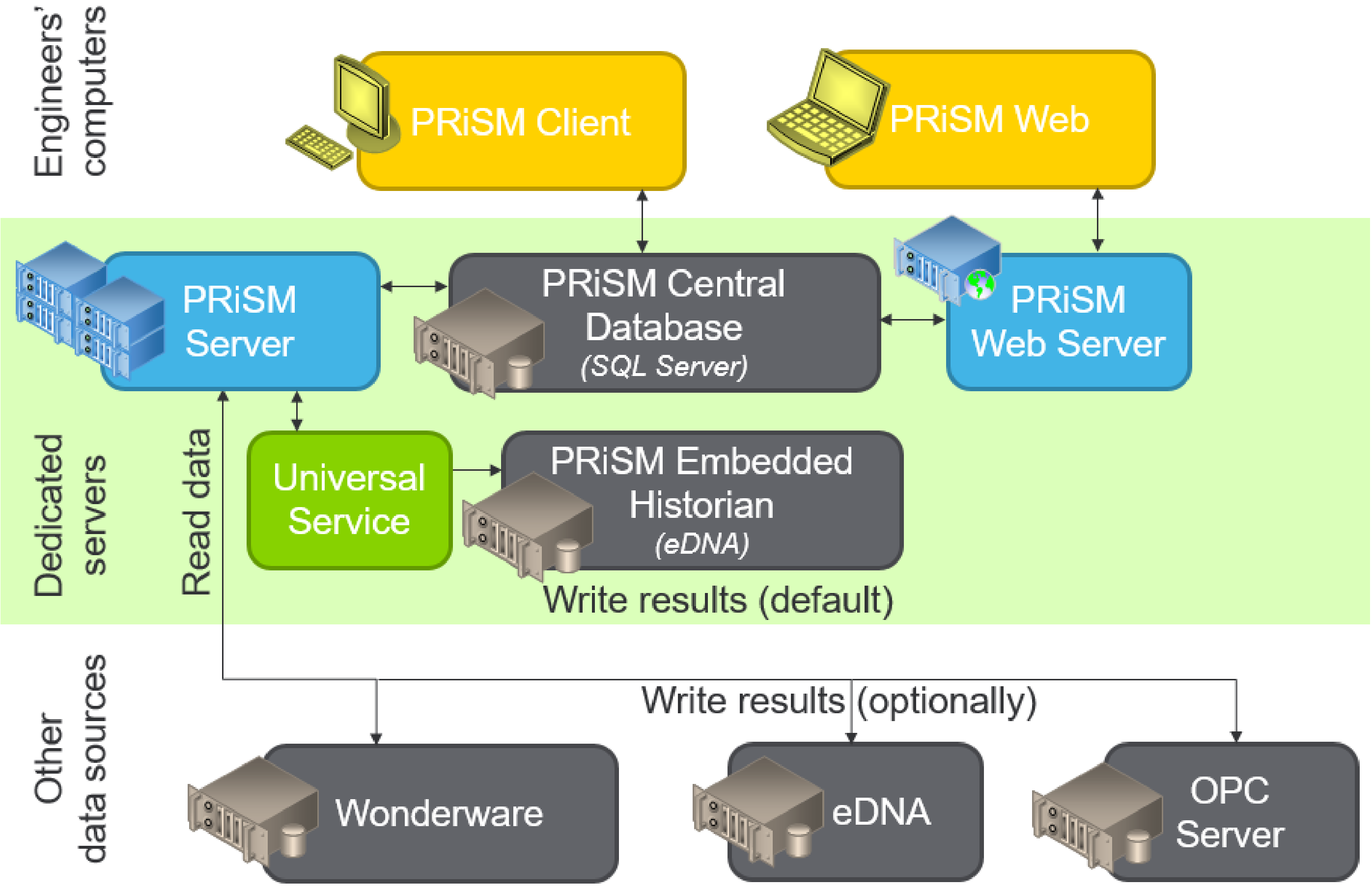
Monitoring philosophy

- Fault diagnostics aren’t always going to be 100% correct but they can be a good starting point
  - Use the “Analysis” button in Web to see what might be happening
  - In addition to the top OMR contributors, also check trends related to any faults that have triggered
  - Always use common sense to verify the correctness of a fault detection



Description	Next Steps
Maintainable Item: Yes Unplanned Shutdown Required: Yes MTTR: Moderate Failure Effect: Bearing Will Fail if No Action is Taken	1. Check that pump is operating within defined operational limits 2. Carry out visual inspection 3. Conduct high speed vibration analysis 4. Plan maintenance intervention to avoid possible failure

Architecture



PRiSM Roadmap & Vision

Subject to change!

Streamlined User Experience

Reduce TCO

Artificial Intelligence

Workforce Empowerment

2019 R1 May 2019

- Prescriptive Information
- Flashless MVP
- Cloud Clients
- Data Quality Management
- Hierarchy Recursive Security

2019 R2 Fall 2019

- Complete HTML5 Web UI
- Data Playback
- Streamlined Installation Program
- Historian Points

2020

- Remaining Useful Life
- Template Automation
- Alerts Prioritization
- Asset Library
- External Access
- External Algorithms

Ideas 2021+

- Batch Analytics
- Central Model Maintenance
- Reinforcement learning

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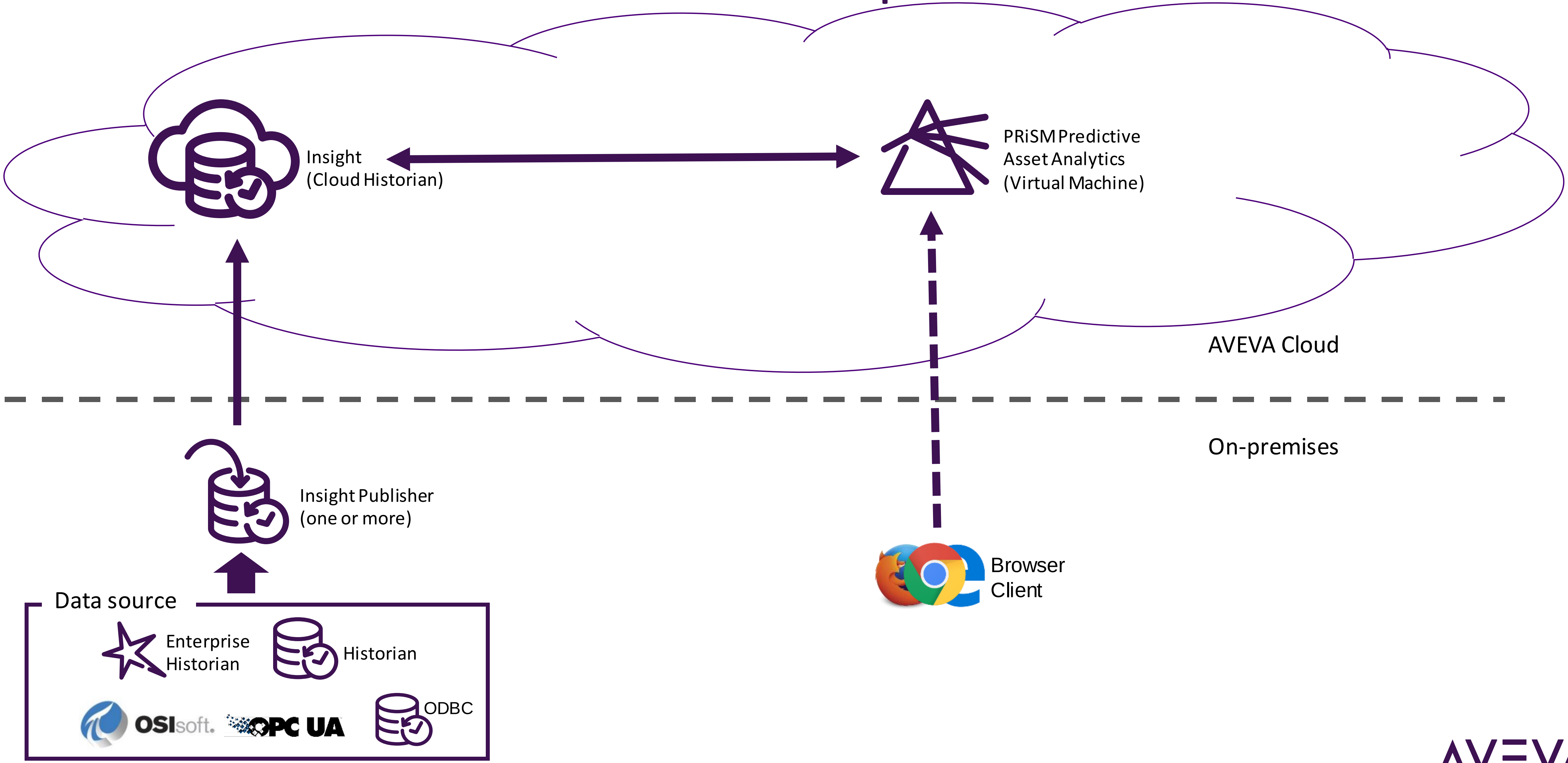
PRiSM in Cloud Hosted Products platform architecture



Questions?

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ABOUT AVEVA

AVEVA is a global leader in engineering and industrial software driving digital transformation across the entire asset and operational life cycle of capital-intensive industries.

The company's engineering, planning and operations, asset performance, and monitoring and control solutions deliver proven results to over 16,000 customers across the globe. Its customers are supported by the largest industrial software ecosystem, including 4,200 partners and 5,700 certified developers. AVEVA is headquartered in Cambridge, UK, with over 4,400 employees at 80 locations in over 40 countries.

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